Excel Product Sales Data Cleaning - Project Report

1. Key Insights

Area	Key findings
Dataset	34 original records
Duplicate records	3 duplicate rows identified
Missing categories	4 records
Missing prices	3 records
Text inconsistencies	Inconsistent capitalization in Product Name
Category errors	Electroni should be Electronics
Data enrichment	Lookup tables used to recover missing categories
Final output	Cleaned and standardized product sales dataset

2. Main Data-Cleaning Results

- Product names standardized using Capitalize Each Word.
- Category spelling corrected from 'Electroni' to *'Electronics'*.
- Extra spaces removed using Trim.
- Four missing categories addressed:

Product Name	Category
Sneakers	Fashion
Coffee Maker	Kitchen
Fitness Tracker	Electronics
Backpack	Outdoor

- Three missing prices handled using average-price imputation.
- Three duplicate records identified for removal.

3. Step-by-Step Project Workflow

1. **Text standardization:** Capitalize product names, correct category spelling, and trim spaces.
2. **Missing category treatment:** In Power Query, create a reference lookup table using Product Name and Category.
3. **Category recovery:** Merge the lookup table using a Left Outer Join and use a Conditional Column to fill missing categories.
4. **Manual category mapping:** Add “Backpack → Outdoor” record because no other backpack record exists.
5. **Missing price treatment:** Use the average price of the same product; if unavailable, use the average price of its category.
6. **Duplicate removal:** Use Data → Remove Duplicates, selecting all columns.
7. **Column splitting:** Extract the Manufacturing Date and Country Code from Product ID.
8. **Column merging:** Combine Product Name and Brand Name into Product_Brand.
9. **Number formatting:** Format Price as Currency and standardize date display where complete date information is available.
10. **Conditional formatting:** Apply Data Bars to Price and highlight Electronics in the Category column.

4. Important Excel Formulas

Column	Formula
Product_Brand	=TEXTJOIN(" ",TRUE,[@[Product Name]],[@[Brand Name]])
Date	=LEFT([@[Product ID]],6)
Manufacturing Date	=DATEVALUE([@Date])
Country Code	=RIGHT([@[Product ID]],2)
Price imputation	=IF(ISBLANK([@[Price ($)]]),IFERROR(AVERAGEIF([Product Name],[@[Product Name]],[Price ($)]),AVERAGEIF([Cleaned_Category],[@Cleaned_Category],[Price ($)])),[@[Price ($)]])

> **Important:** Product ID has Day, Month and Country but no Year. DATEVALUE assumes the current year (2026), so the year is an assumption.

5. Imputed Price Values

Product Name	Method	Price	On Basis
Headphones	Same product	$250	Headphones Bose
Sunglasses	Same product	$150	Sunglasses Oakley
Camping Tent	Category average	$100	Outdoor: $70 and $130

6. Final Data Transformation

Original column	Final column
Product ID	Manufacturing Date
Product ID	Country Code
Product Name + Brand Name	Product_Brand
Category	Cleaned_Category
Price ($)	Currency-formatted Price

7. Conditional Formatting

- **Price:** Apply Blue Data Bars to compare product prices visually.
- **Category:** Highlight Electronics using the formula:

```
=$J2="Electronics"
```

8. Key Recommendations

- Maintain separate lookup tables for future missing-category corrections.
- Avoid replacing missing prices with zero, as this can distort sales analysis.
- Keep a record of manually assigned categories.
- Validate duplicate removal using all columns.
- Add a proper year field before converting manufacturing date codes into complete dates.
- Refresh Power Query after updating the source data.

9. Future Scope

The cleaned dataset provides a reliable foundation. The following enhancements can extend this project into a more automated and analytical workflow.

- **Add a year field:** Include a full date or year in the source data so manufacturing dates no longer rely on the assumed year (2026).
- **Sales and revenue analysis:** Calculate Revenue (Price × Quantity) and build PivotTables and dashboards by category, brand, country and month.
- **Automated refresh pipeline:** Extend the Power Query workflow so new sales records are cleaned automatically on refresh, with no manual steps.
- **Data validation controls:** Add drop-down lists for Category and Country Code, and input rules, to prevent misspellings such as “Electroni” at the source.
- **Master lookup tables:** Maintain central Product-to-Category and Brand reference tables so missing categories are recovered consistently, including manual mappings such as Backpack → Outdoor.
- **Advanced imputation:** Test median or regression-based price imputation, and add a flag column marking which prices were imputed.
- **Outlier and quality checks:** Add automated checks for unusual prices (for example, Blender prices range from $90 to $400) and for duplicate or invalid Product IDs.
- **Country mapping:** Map country codes to country names and regions to support geographic reporting.
- **Scaling to larger data:** Reproduce the same cleaning rules in Python (pandas) or SQL when the dataset outgrows Excel.

10. Sheet 1: Raw Dataset

The original, uncleaned product sales data with 34 records and 6 columns (Product ID, Product Name, Brand Name, Price, Quantity, Category).

Highlights: red = missing value; orange = misspelled category ("Electroni"); purple = repeated record (duplicate of an earlier row).

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	laptop	Dell	$1,000	30	Electronics
15-FEB-US	Sneakers	Nike	$80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	$130	40	Kitchen
11-APR-US	smartphone	Samsung	$900	25	Electronics
22-MAY-US	Backpack	North Face	$70	20	
07-JUN-UK	Headphones	Sony		45	Electroni
19-JUL-UK	T-shirt	Adidas	$30	5	Fashion
23-AUG-UK	Blender	Ninja	$90	35	Kitchen
05-SEP-UK	Tablet	Apple	$500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	$130	10	Outdoor
17-JUN-IN	Laptop	HP	$950	25	Electronics
25-NOV-AU	Sneakers	Adidas	$90	40	
08-DEC-DE	Coffee Maker	Nespresso	$120	35	
18-FEB-CA	Smartwatch	Fitbit	$150	15	Electroni
16-APR-ES	headphones	Bose	$250	20	Electroni
21-AUG-CA	Laptop Bag	Samsonite	$50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	$160	15	Electroni
27-JAN-IT	laptop	Asus	$980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	$150	15	Fashion
14-AUG-US	Camping Tent	Coleman		10	Outdoor
14-MAY-RU	Camera	Nikon	$700	50	Electronics
09-JAN-CA	Microwave	Panasonic	$80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	$150	30	
21-AUG-CA	Laptop Bag	Samsonite	$50	35	Accessories
29-SEP-CA	Smartphone	Google	$800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion
11-JUL-CA	Blender	Vitamix	$400	40	Kitchen
16-APR-ES	headphones	Bose	$250	20	Electroni
07-MAR-CA	Dress	Zara	$60	30	Fashion
13-APR-CA	Toaster	Hamilton	$40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	$130	5	Electronics
02-DEC-CA	Jeans	Levi's	$50	50	Fashion
17-JUN-IN	Laptop	HP	$950	25	Electronics
09-JUL-FR	Watch	Casio	$100	20	Accessories

11. Sheet 2: Data Cleaning & Imputation

The working sheet where the cleaning steps were applied: duplicates removed (34 to 31 records), text standardized, Product ID split into Date, Manufacturing Date and Country Code, Product_Brand created, and missing prices and categories filled (Headphones $250, Camping Tent $100, Sunglasses $150).

Conditional formatting (as in Excel): Blue Data Bars on Cleaned_Price ($) and Electronics highlighted in the Cleaned_Category column.

Product ID	Date	Manufacturing Date	Country Code	Product Name	Brand Name	Product_Brand	Price ($)	Cleaned_Price ($)	Quantity	Cleaned_Category
28-JAN-US	28-JAN	28-Jan-2026	US	Laptop	Dell	Laptop Dell	1000	$1,000	30	Electronics
17-JUN-IN	17-JUN	17-Jun-2026	IN	Laptop	HP	Laptop HP	950	$950	25	Electronics
27-JAN-IT	27-JAN	27-Jan-2026	IT	Laptop	Asus	Laptop Asus	980	$980	10	Electronics
15-FEB-US	15-FEB	15-Feb-2026	US	Sneakers	Nike	Sneakers Nike	80	$80	15	Fashion
25-NOV-AU	25-NOV	25-Nov-2026	AU	Sneakers	Adidas	Sneakers Adidas	90	$90	40	Fashion
03-MAR-US	03-MAR	03-Mar-2026	US	Coffee Maker	Keurig	Coffee Maker Keurig	130	$130	40	Kitchen
08-DEC-DE	08-DEC	08-Dec-2026	DE	Coffee Maker	Nespresso	Coffee Maker Nespresso	120	$120	35	Kitchen
11-APR-US	11-APR	11-Apr-2026	US	Smartphone	Samsung	Smartphone Samsung	900	$900	25	Electronics
22-MAY-US	22-MAY	22-May-2026	US	Backpack	North Face	Backpack North Face	70	$70	20	Outdoor
07-JUN-UK	07-JUN	07-Jun-2026	UK	Headphones	Sony	Headphones Sony		$250	45	Electronics
16-APR-ES	16-APR	16-Apr-2026	ES	Headphones	Bose	Headphones Bose	250	$250	20	Electronics
19-JUL-UK	19-JUL	19-Jul-2026	UK	T-Shirt	Adidas	T-Shirt Adidas	30	$30	5	Fashion
23-AUG-UK	23-AUG	23-Aug-2026	UK	Blender	Ninja	Blender Ninja	90	$90	35	Kitchen
05-SEP-UK	05-SEP	05-Sep-2026	UK	Tablet	Apple	Tablet Apple	500	$500	50	Electronics
14-OCT-UK	14-OCT	14-Oct-2026	UK	Hiking Boots	Timberland	Hiking Boots Timberland	130	$130	10	Outdoor
18-FEB-CA	18-FEB	18-Feb-2026	CA	Smartwatch	Fitbit	Smartwatch Fitbit	150	$150	15	Electronics
20-AUG-CN	20-AUG	20-Aug-2026	CN	Smartwatch	Huawei	Smartwatch Huawei	160	$160	15	Electronics
21-AUG-CA	21-AUG	21-Aug-2026	CA	Laptop Bag	Samsonite	Laptop Bag Samsonite	50	$50	35	Accessories
01-MAR-UK	01-MAR	01-Mar-2026	UK	Sunglasses	Oakley	Sunglasses Oakley	150	$150	15	Fashion
14-AUG-US	14-AUG	14-Aug-2026	US	Camping Tent	Coleman	Camping Tent Coleman		$100	10	Outdoor
14-MAY-RU	14-MAY	14-May-2026	RU	Camera	Nikon	Camera Nikon	700	$700	50	Electronics
09-JAN-CA	09-JAN	09-Jan-2026	CA	Microwave	Panasonic	Microwave Panasonic	80	$80	20	Kitchen
19-JUL-BR	19-JUL	19-Jul-2026	BR	Fitness Tracker	Xiaomi	Fitness Tracker Xiaomi	150	$150	30	Electronics
29-SEP-CA	29-SEP	29-Sep-2026	CA	Smartphone	Google	Smartphone Google	800	$800	45	Electronics
03-JUN-CA	03-JUN	03-Jun-2026	CA	Sunglasses	Ray-Ban	Sunglasses Ray-Ban		$150	25	Fashion
11-JUL-CA	11-JUL	11-Jul-2026	CA	Blender	Vitamix	Blender Vitamix	400	$400	40	Kitchen
07-MAR-CA	07-MAR	07-Mar-2026	CA	Dress	Zara	Dress Zara	60	$60	30	Fashion
13-APR-CA	13-APR	13-Apr-2026	CA	Toaster	Hamilton	Toaster Hamilton	40	$40	10	Kitchen
24-MAY-CA	24-MAY	24-May-2026	CA	Fitness Tracker	Garmin	Fitness Tracker Garmin	130	$130	5	Electronics
02-DEC-CA	02-DEC	02-Dec-2026	CA	Jeans	Levi's	Jeans Levi's	50	$50	50	Fashion
09-JUL-FR	09-JUL	09-Jul-2026	FR	Watch	Casio	Watch Casio	100	$100	20	Accessories

12. Sheet 3: Final Cleaned Data

The final, analysis-ready dataset with 31 unique records and 9 columns, with standardized product names, corrected categories, imputed prices and currency-formatted Price.

Conditional formatting (as in Excel): Blue Data Bars on Price ($) and Electronics highlighted in the Category column.

Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Product_Brand	Price ($)	Quantity	Category
28-JAN-US	28-Jan-2026	US	Laptop	Dell	Laptop Dell	$1,000	30	***Electronics***
17-JUN-IN	17-Jun-2026	IN	Laptop	HP	Laptop HP	$950	25	***Electronics***
27-JAN-IT	27-Jan-2026	IT	Laptop	Asus	Laptop Asus	$980	10	***Electronics***
15-FEB-US	15-Feb-2026	US	Sneakers	Nike	Sneakers Nike	$80	15	Fashion
25-NOV-AU	25-Nov-2026	AU	Sneakers	Adidas	Sneakers Adidas	$90	40	Fashion
03-MAR-US	03-Mar-2026	US	Coffee Maker	Keurig	Coffee Maker Keurig	$130	40	Kitchen
08-DEC-DE	08-Dec-2026	DE	Coffee Maker	Nespresso	Coffee Maker Nespresso	$120	35	Kitchen
11-APR-US	11-Apr-2026	US	Smartphone	Samsung	Smartphone Samsung	$900	25	***Electronics***
22-MAY-US	22-May-2026	US	Backpack	North Face	Backpack North Face	$70	20	Outdoor
07-JUN-UK	07-Jun-2026	UK	Headphones	Sony	Headphones Sony	$250	45	***Electronics***
16-APR-ES	16-Apr-2026	ES	Headphones	Bose	Headphones Bose	$250	20	***Electronics***
19-JUL-UK	19-Jul-2026	UK	T-Shirt	Adidas	T-Shirt Adidas	$30	5	Fashion
23-AUG-UK	23-Aug-2026	UK	Blender	Ninja	Blender Ninja	$90	35	Kitchen
05-SEP-UK	05-Sep-2026	UK	Tablet	Apple	Tablet Apple	$500	50	***Electronics***
14-OCT-UK	14-Oct-2026	UK	Hiking Boots	Timberland	Hiking Boots Timberland	$130	10	Outdoor
18-FEB-CA	18-Feb-2026	CA	Smartwatch	Fitbit	Smartwatch Fitbit	$150	15	***Electronics***
20-AUG-CN	20-Aug-2026	CN	Smartwatch	Huawei	Smartwatch Huawei	$160	15	***Electronics***
21-AUG-CA	21-Aug-2026	CA	Laptop Bag	Samsonite	Laptop Bag Samsonite	$50	35	Accessories
01-MAR-UK	01-Mar-2026	UK	Sunglasses	Oakley	Sunglasses Oakley	$150	15	Fashion
14-AUG-US	14-Aug-2026	US	Camping Tent	Coleman	Camping Tent Coleman	$100	10	Outdoor
14-MAY-RU	14-May-2026	RU	Camera	Nikon	Camera Nikon	$700	50	***Electronics***
09-JAN-CA	09-Jan-2026	CA	Microwave	Panasonic	Microwave Panasonic	$80	20	Kitchen
19-JUL-BR	19-Jul-2026	BR	Fitness Tracker	Xiaomi	Fitness Tracker Xiaomi	$150	30	***Electronics***
29-SEP-CA	29-Sep-2026	CA	Smartphone	Google	Smartphone Google	$800	45	***Electronics***
03-JUN-CA	03-Jun-2026	CA	Sunglasses	Ray-Ban	Sunglasses Ray-Ban	$150	25	Fashion
11-JUL-CA	11-Jul-2026	CA	Blender	Vitamix	Blender Vitamix	$400	40	Kitchen
07-MAR-CA	07-Mar-2026	CA	Dress	Zara	Dress Zara	$60	30	Fashion
13-APR-CA	13-Apr-2026	CA	Toaster	Hamilton	Toaster Hamilton	$40	10	Kitchen
24-MAY-CA	24-May-2026	CA	Fitness Tracker	Garmin	Fitness Tracker Garmin	$130	5	***Electronics***
02-DEC-CA	02-Dec-2026	CA	Jeans	Levi's	Jeans Levi's	$50	50	Fashion
09-JUL-FR	09-Jul-2026	FR	Watch	Casio	Watch Casio	$100	20	Accessories

13. Project Conclusion

- The project demonstrates how Excel and Power Query can transform inconsistent raw sales data into a cleaner, more reliable dataset.
- The process improves data consistency, handles missing information, removes duplicate records, standardizes product descriptions, and prepares the dataset for further analysis and reporting.